Department of Applied Mathematics
and
Computational Sciences

PSG College of Technology

Coimbatore

Course File
Of
18XTE3 NATURAL LANGUAGE PROCESSING

Name of the Faculty	SREEJA NK
Course period	2020-2021 ODD SEMESTER
Semester no	5
Email id	nksreeja21@gmail.com
Contact No	9659425507

P S G College of Technology

Vision

PSG College of Technology aspires to be recognised as one of the leaders in engineering education, research and application of knowledge to benefit society.

Mission

Provide world-class Engineering Education, Foster Research and Development. Evolve innovative applications of Technology. Encourage Entrepreneurship. Ultimately mould young men and women capable of assuming leadership of the society for the betterment of the Country.

Department of Applied Mathematics
And
Computational Sciences

Vision

"Stay ahead and be relevant."

Mission

The fundamental objective of the department is to develop quality professionals by providing concept oriented subject knowledge through high quality teaching supplemented with practical training. Apart from specialized knowledge and skills, the programmes conducted by the Department aim to develop the personality of students by inculcating values of honesty, sincerity, team spirit and work culture.

M.Sc (Theoretical Computer Science)

Programme Educational Objectives

PEO1	To motivate the graduates of the programme to obtain opportunities for research and pursue higher studies in reputed universities.
PEO2	To provide a foundation for immediate employability in top-notch industries and government organizations.
PEO3	To encourage the students to become entrepreneurs and contribute to the society.
PEO4	To provide value systems to the students to become a responsible citizen.

Programme Outcome

PO1	An ability to apply the theoretical knowledge of Mathematics and Computational Sciences to model and solve real time problems.
PO2	An ability to understand, analyze and design efficient algorithms.
PO3	Development of soft skills and practicing professional ethics
PO4	An ability to design efficient protocols for advanced communication technology.
PO5	Acquire knowledge of contemporary computational issues.
PO6	In depth knowledge of foundations of computing.
PO7	An ability to understand and solve emerging research problems.
PO8	Develop programming skills to implement research projects.

18XTE3 NATURAL LANGUAGE PROCESSING

Course objectives		Course outcomes		Number of hours	Related program outcomes
1	To introduce the challenges of empirical methods for natural language processing (NLP) applications	CO1 **LOT**	Students will **understand** the basic concepts and the analysis levels in NLP. Students will understand the importance of language modelling.	5	PO01, PO02
2	To make students understand the techniques used in morphological and lexical analysis.	CO2 **LOT**	Students will **understand** the construction of a Finite state transducer for morphological analysis and various algorithms used for lexical analysis. They will be able to **create** FST for a given problem	12	PO01, PO02, PO05, PO06
3	To make students understand various parsing methods used for NLP.	CO3 **HOT**	Students will be able to understand and **apply** various parsing methods and to estimate probabilistic model parameters during syntactic analysis.	12	PO01, PO02, PO05, PO06, PO07, PO08
4	To make students understand the methods used to analyze the language at semantic and pragmatic level.	CO4 **HOT**	Students will be able to **apply** proper algorithms to **analyse** the semantics of the language and methods to resolve ambiguity.	12	PO01, PO02, PO05, PO06, PO07, PO08
5	To make students understand algorithms behind the generation of a natural language and applications of NLP	CO5 **HOT**	Students will be able to select and **apply** various algorithms for sentiment analysis and apply the knowledge to solve new research problems.	4	PO01, PO02, PO05, PO06, PO07, PO08

INTRODUCTION : Natural language processing techniques **-** The different analysis levels used for NLP: morpho - lexical - syntactic – semantic - pragmatic - markup (TEI, UNICODE) – Applications – open problems. (3)

WORDS : Regular expressions – Automata – Morphology – Finite state Transducers – Finite state morphological parsing – Combining FST lexicon and rules – Porter Stemmer Algorithm – Probabilistic models for Spelling – Bayes method, Minimum edit distance - N-Grams – Counting words in Corpora – Simple n-grams – Smoothing – Evaluating language models : Entropy, Perplexity- Part of Speech Tagging (POS) – Rule based tagging – Stochastic based tagging – Transformation based tagging - Context Free Grammars - Top down parser – Earley Algorithm – Bottom-up parsing – CYK parser – Probabilistic parsing. (15)

SEMANTICS : Meaning structure of a language – First order predicate calculus – Alternative approaches to meaning – Syntax driven sematic analysis – Attachments for a fragment of English – Word Sense Disambiguation – Machine learning approaches – Dictionary based approaches – Pragmatics : Discourse – Text coherence (10)

Natural Language Generation : Architecture of NLG Systems- Generation Tasks and Representations- Application of NLG. Machine Translation: Language similarities and differences – The transfer metaphor – Direct translation – Statistical translation - Translation involving Indian Languages. (12)

CASE STUDIES : Sentiment Analysis **-** Automatic summarization - Information retrieval and Question answering - Named entity recognition and relation extraction – Deep Learning for NLP (5)

TUTORIAL PRACTICE:

1. Implementing word similarity.
2. Implementing simple problems related to word disambiguation.
3. Simple demonstration of part of speech tagging.
4. Translation from one language into another language
5. Word sense disambiguation
6. Representations of Text, if, if-idf, WordZVec, Classifications Clustering of Text Data.

Total L:45+T:30 = 75

TEXT BOOKS:

1. Daniel Jurafsky and James H. Martin, "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition", Prentice Hall, 2014.
2. Christopher Manning and Hinrich Schütze, "Foundations of Statistical Natural Language Processing", MIT Press, 2003.

REFERENCES:

1. James Allen, "Natural Language Understanding", Addison Wesley, 2008.
2. Steven Bird, Ewan Klein, and Edward Loper, "Natural Language Processing with Python - Analyzing Text with the Natural Language Toolkit", O'Reilly Media, 2009.

DETAILED COURSE PLAN

WEEK	TOPICS
1	**INTRODUCTION :** Natural language processing techniques **-** The different analysis levels used for NLP: morpho - lexical - syntactic – semantic - pragmatic - markup (TEI, UNICODE) – Applications – open problems.
2	**WORDS :** Regular expressions – Automata – Morphology – Finite state Transducers – Finite state morphological parsing
3	Combining FST lexicon and rules – Porter Stemmer Algorithm – Probabilistic models for Spelling – Bayes method, Minimum edit distance
4	N-Grams – Counting words in Corpora – Simple n-grams – Smoothing .Evaluating language models : Entropy, Perplexity
5	Part of Speech Tagging (POS) – Rule based tagging – Stochastic based tagging – Transformation based tagging
6	Context Free Grammars - Top down parser – Earley Algorithm – Bottom-up parsing – CYK parser – Probabilistic parsing.
7	**SEMANTICS** : Meaning structure of a language – First order predicate calculus – Alternative approaches to meaning
8	Syntax driven sematic analysis – Attachments for a fragment of English – Word Sense Disambiguation – Machine learning approaches
	TEST I
9	Dictionary based approaches – Pragmatics : Discourse – Text coherence
10	**Natural Language Generation** : Architecture of NLG Systems- Generation Tasks and Representations
11	Application of NLG. Machine Translation: Language similarities and differences
12	The transfer metaphor – Direct translation
13	Statistical translation - Translation involving Indian Languages.
	TEST II
14	**CASE STUDIES :** Sentiment Analysis **-** Automatic summarization - Information retrieval and Question answering
15	Named entity recognition and relation extraction

16	Deep Learning for NLP
17	Assignment presentation
18	Assignment presentation
19	Assignment presentation

Relationship between Course Outcomes (COs) and Programme Outcomes (POs)

Course Outcome	Programme Outcomes							
	1	2	3	4	5	6	7	8
CO1	*	*						
CO2	*	*			*	*		
CO3	*	*			*	*	*	*
CO4	*	*			*	*	*	*
CO5	*	*			*	*	*	*

Unit-wise Scope for Outcome and Bloom's Taxonomy

CO / Blooms Taxonomy	CO1	CO2	CO3	CO4	CO5	Relative Frequency of Scope (RF_i)	Weight W_i
Creating		*		*	*	3/5	6
Evaluating				*	*	2/5	5
Analysing		*	*	*	*	4/5	4
Applying			*	*	*	3/5	3
Understanding	*	*	*	*		4/5	2
Remembering	*	*	*			3/5	1

<u>BTI = Bloom's Taxonomy Index:</u> $\frac{\sum RF_i W_i}{\sum W_i}$ = 0.6095

<u>Level of the Course:</u> HoT (Hint: BTI ≥ 0.5 HoT (Higher order Thinking),
BTI < 0.5 LoT (Lower order Thinking)